

Manaswi Saha, Ph. D.

HCI Research Scientist | Human-AI Interaction | Emerging Tech Design & Development

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EDUCATION

University of Washington, Seattle

MS & Ph.D. Computer Science & Engg.

📅 09/2017 - 08/2022 📍 Seattle, WA

- Google PhD Fellowship 2020-22 • 2019 Amazon Catalyst Award \$10K • CHI'19 Best Paper Award

University of Maryland (transferred to University of Washington)

Ph.D. in Computer Science

📅 08/2015 - 08/2017 📍 College Park, MD

- Dean's Fellowship 2015 - 2017

Vellore Institute of Technology University

Master of Computer Applications (MCA)

📅 07/2009 - 05/2012 📍 Vellore, India

- Merit Scholarship: awarded all three years – top 3 (of 120) students

University of Mumbai

Bachelor of Science Information Technology

📅 06/2006 - 04/2009 📍 Mumbai, India

- Certificate of Merit: awarded all three years – top 3 (of 60) students

SKILLS

Research Dev: system building • design space development • public deployments

Programming: Python, C, C++, Java | Front-end: HTML/CSS, JavaScript, Play, Django, React, Figma • mapbox, d3, kepler.gl | Backend: PostgreSQL, MySQL, MongoDB | Others: Git, Amazon Mechanical Turk, Prolific

Applied AI/ML Development

LLM Dev: Cursor, Gemini, v0, HuggingFace • Data Analysis: Python – pandas, numpy, sklearn

User Research

domain-specific problem discovery and understanding, user studies, semi-structured interviews, surveys, affinity diagramming, stakeholder analysis, thematic analysis

Communication and Leadership

cross-functional team management and collaborations, C-suite stakeholder communication, people development, student mentorship

SUMMARY

An Applied R&D Scientist with expertise in human-centered technology design and development and multi-disciplinary research and 12+ years of experience, leveraging emerging technologies for building innovative AI-driven immersive systems and human experiences.

Research Areas: HCI • Human-AI Interaction • AR • Future of Work • Urban Informatics • Accessibility • Sustainability • Data Visualization

Technical Areas of Interest: Human-AI/agent interaction and experiences • AI assistants — multimodal conversational AI, always-on contextual AI, emotion AI • AR — visual, audio

INDUSTRY EXPERIENCE



Accenture Labs • Digital Experiences Group

Associate Principal Researcher

📅 09/2022 - Present 📍 San Francisco, CA

Team Leads/Managers: Alex Kass, Mirjana Spasojevic, and Mike Kuniavsky

- **Leading HCI research, focusing on Human-AI Interaction** to inform enterprise applications: designing AI/AR-assisted workflows and pipelines for diverse worker profiles and industry verticals (e.g., utilities, healthcare, media & entertainment); exploring multimodal conversational AI agents, combining biosensing and applied neuroscience for various business use cases (e.g., workforce training, filmmaking)
- **Designing and conducting user studies, developing technical prototypes, and translating research** into business value for \$1B+ clients, through presentations and reports, informing innovation and strategic decisions
- **Leading research for AI-driven real-time tools for cross-disciplinary knowledge support** as the Principal Investigator, involving development and testing of LLM-based prototypes for diverse use cases (e.g., video learning, online meetings, scientific reading) and publishing across multiple academic publications (CHI, CSCW)
- **Spearheading Audio-AR research**, developing conversational AI for physical tasks, collaborating with Cornell Tech on healthcare applications; paper at CUI
- **Mentored and managed 9 PhD and undergraduate students** during summer internships and university collaborations across various research initiatives
- **Managing the team's graduate and undergraduate internship program**, including developing new streamlined processes for hiring as well as internship experience, ensuring uniform experience across remote and in-person interns



Autodesk Research • HCI/VIS: User Interface Research Group

Research Intern

📅 06/2020 - 10/2020 📍 (Remote) Toronto, Canada

Mentor: Justin Matejka

- Studied people's estimation skills for metrics (e.g., cost, power, carbon footprint) for insights on effective communication of sustainability data and metrics
- Built a HTML/CSS/JS app and ran an online study with 50+ sustainability experts and novices
- Communicated study results to research leadership and broader internal audience to demonstrate the implications for future product enhancements such as new sustainability features for CAD software



Microsoft Research • Ability and Enable Groups

Research Intern

📅 2018 - 09/2018 📍 Redmond, WA

Mentors: Meredith R. Morris, Ed Cutrell, Alex Fiannaca

- Engineered Landmark AI, a mobile app prototype using 3D-audio and computer vision algorithms for addressing the last-few-meters challenge in GPS systems
- Conducted a design-probe study to understand and address wayfinding challenges using GPS tools for visually impaired users: white cane and guide dog users
- Ran a survey and an interview study with 12 participants to create the design space for AI-based navigation tools catering people with visual disabilities
- Published results in ASSETS 2019

SELECTED PUBLICATIONS

-  Steering AI-Driven Personalization of Scientific Text for General Audiences • ACM CSCW 2025
T. Kim, D. Agarwal, J. Ackerman, **M. Saha**
-  Exploring the Design Space of Real-time LLM Knowledge Support Systems: A Case Study of Jargon Explanations • ACM CHI 2025
Y. Liu, A. Shah, J. Ackerman, **M Saha**
-  Situated Conversational Agents for Task Guidance: A Preliminary User Study • ACM CUI 2024
AWD Bremers*, **M Saha***, AG Ramirez-Aristizabal
-  Visualizing Urban Accessibility: Investigating Multi-stakeholder Perspectives through a Map-based Design Probe Study • CHI 2022
M. Saha, D. Chauhan, S. Patil, R. Kangas, J. Heer, and J. E. Froehlich
-  Urban Accessibility as a Socio-Political Problem: A Multi-Stakeholder Analysis • CSCW 2020
M. Saha, A. J. Fiannaca, M. Kneisel, E. Cutrell, M. R. Morris
-  Closing the Gap: Designing for the Last-Few-Meters Wayfinding Problem for People with Visual Impairments • ASSETS 2019
M. Saha, A. J. Fiannaca, M. Kneisel, E. Cutrell, M. R. Morris
-  Project Sidewalk: A Web-based Crowdsourcing Tool for Collecting Sidewalk Accessibility Data at Scale • CHI 2019 • Best Paper (Top 1%)
M. Saha, M. Saugstad, H. Maddali, A. Zeng, R. Holland, S. Bower, A. Dash, S. Chen, A. Li, K. Hara, J. Froehlich

ACCOMPLISHMENTS

-  Engineered Project Sidewalk, a tool deployed in 20+ cities around the world; used by 15000+ users; generated 1M+ point datasets
-  Research recognized by media, notably in New York Times, NPR, and local TV news in Seattle and DC
-  Invited speaker across academic institutions, tech industry, government conferences, and local meetups

INDUSTRY EXPERIENCE



Adobe Research • Big Data Experience Lab

Research Intern

05/2016 - 08/2016

San Jose, CA

Mentors: Tom Jacobs and David Tompkins

- Created an ecosystem design and built prototype for Bluetooth beacon-based personalized information delivery system for digital marketing that bridges the online world with the physical (brick and mortar stores)
- Submitted Patent as the lead

ACADEMIC EXPERIENCE

University of Washington

Graduate Assistant • Makeability Lab

09/2017 - 08/2022

Seattle, WA

Advisor: Jon Froehlich • **Collaborator:** Jeffrey Heer

- Led 4—10-person design, engineering, and research teams over 6 years
- Lead engineer and researcher for making the built infrastructure (e.g., sidewalks) and tools accessible to individuals with diverse abilities and capabilities
- Wrote 15.5K lines of code for HTML/CSS/JS frontend and Java/Scala/PostgreSQL backend
- Engineered tools and prototypes for collecting and mapping accessibility data at scale (e.g., Project Sidewalk); used Google Street View, crowdsourcing, gamification, interactive geovisualizations, AI, and human-centered design techniques
- Designed and executed interview studies with 35+ participants to understand multi-stakeholder data-driven decision-making needs and sensemaking practices around urban accessibility; stakeholders included policymakers, disability advocates, government officials, people with mobility disabilities, and caregivers
- Evaluated tools using lab studies, public deployments in 5+ cities, and user interviews with varied stakeholders
- Impact: Project Sidewalk deployed by 20+ cities around the globe, generated 1M+ point sidewalk accessibility datasets, and used by city governments and NGOs
- Paper(s): CHI, CSCW, VIS, and ASSETS

University of Maryland

Graduate Assistant • Makeability Lab

08/2015 - 08/2017

College Park, MD

Advisor: Jon Froehlich

- Explored the use of thermal cameras mounted on smartphones by novices (e.g., DIY enthusiasts) to conduct thermography in homes
- Analyzed interviews for a 4-week field study with 10 participants
- Paper(s): CHI 2016 and 2017

IIIT-Delhi

Research Associate • Mobile & Ubiquitous Comp.

11/2012 - 07/2015

Delhi, India

Advisor: Amarjeet Singh • **Collaborators:** Anind Dey (UW), Yuvraj Agarwal (CMU), Pushpendra Singh

- Led engineering for SensorAct, a building Java/MongoDb-based middleware to control and operate hardware sensor modules, mounted with ambient environmental sensors, through an online interface for building managers and occupants
- Engineered EnergyLens+, a real-time energy apportionment and feedback system for smart living spaces that interfaced with smartphone sensors and smart meters for inferring and apportioning personal energy consuming activities to individuals
- Engineered the end-to-end system using Python/Django/MySQL server and designed an Android app for visualizing feedback
- Ran a 2-week field study to evaluate the system with a small-scale deployment in single and multi-occupant homes
- Mentored an undergraduate student for Android app development, now a NYU professor
- Paper(s): ACM eEnergy, BuildSys, IEEE UIC, and NSDI

TEACHING

Human-Centered Computing Courses at University of Washington

Intro to HCI • Advanced HCI • The Future of Access Technologies • Crowdsourcing, Citizen Science, and Large-scale Online Experimentation • Capstone Software Design to Empower Underserved Populations

Programming at University of Maryland

Object-Oriented Programming I • Object-Oriented Programming II